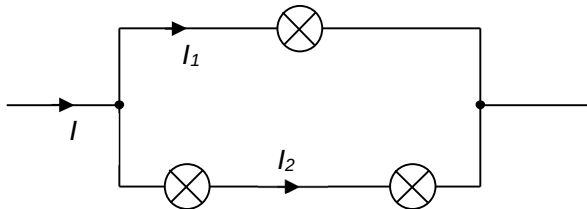


20 Maximum power for circuit can be attained when the single bulb is at maximum power.



$$I_2 = \frac{1}{2} I_1$$

Hence the power for each bulb on the lower branch will be $\frac{1}{4}$ maximum power, which is 2.5 W.

Maximum power for circuit = $10 + 2.5 + 2.5 = 15$ W

Note: If each bulb on lower branch produces maximum power, the bulb on the upper branch will blow (more than max. power produced).

Answer: B

21 Current flowing in opposite direction and as Force by M on N is towards the right